

Business Sales Performance Analytics

1. Project Objective

To analyse business sales data and identify sales trends, profitable categories, regional performance, and loss-making products in order to support data-driven business decisions and recommend growth strategies.

2. Problem Statement

The business aims to understand its sales performance and address the following key questions:

1. Which products and categories generate the highest revenue and profit?
 2. How have sales and profits changed over time?
 3. Which regions contribute the most to business growth?
 4. Which products or sub-categories are causing losses?
 5. What strategies can be implemented to improve profitability and accelerate business growth?
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4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
Oder ID	Text	Unique order identification number
Order Date	Short Date	Date on which the order was placed
Ship Date	Short Date	Date on which the order was shipped
Customer ID	Text	Unique identifier of customers
Customer Name	Text	Name of the customers
Country	Text	Country where customers belong
City	Text	Customer city location
State	Text	Customer state location
Postal Code	Number	Postal code of customer location
Region	Text	Business region (West, East, Central, South)
Product ID	Text	Unique identifier of products
Category	Text	Product category

Sub-Category	Text	Product sub-category
Product Name	Text	Name of the product
Sales	Currency	Revenue generated from product sales
Quantity	Number	Number of units sold
Discount	Percentage	Discount provided on sales
Profit	Currency	Profit earned from each transaction
Year	Number	Extracted year from order date
Month	Text	Extracted month from order date
Quarter	Text	Extracted quarter from order date

5. Tools & Technologies

- **Python (Google Colab):** Data cleaning, preprocessing, exploratory data analysis (EDA), and business insight generation.
- **Pandas:** Data manipulation, transformation, and statistical analysis.
- **Matplotlib & Seaborn:** Data visualization and trend analysis.
- **Power BI:** DAX calculations, interactive visualizations, and dashboard development.
- **Microsoft Excel / CSV:** Dataset storage and export of cleaned data for reporting and dashboard integration.
- **GitHub:** Project version control, documentation, and portfolio presentation.

6. Data pre-processing

The following data preprocessing steps were performed to ensure data quality and prepare the dataset for analysis:

1. Data Loading

- Imported the dataset into Google Colab using Pandas.

2. Duplicate Handling

- Identified duplicate records using `df.duplicated().sum()`.
- Removed duplicate rows using `df.drop_duplicates()`.

3. Missing Value Analysis

- Checked for null values using `df.isnull().sum()`.

- Verified data completeness before proceeding with analysis.

4. Date Format Standardization

- Converted mixed date formats into proper datetime format using `pd.to_datetime()`.
- Extracted additional time-based features such as Year, Month, and Quarter for trend analysis.

5. Data Type Validation

- Verified and corrected data types of numerical, categorical, and date columns using `df.info()`.

6. Feature Engineering

- Created additional columns such as:
 1. Year
 2. Month
 3. Quarter
- These features were used for time-series and trend analysis.

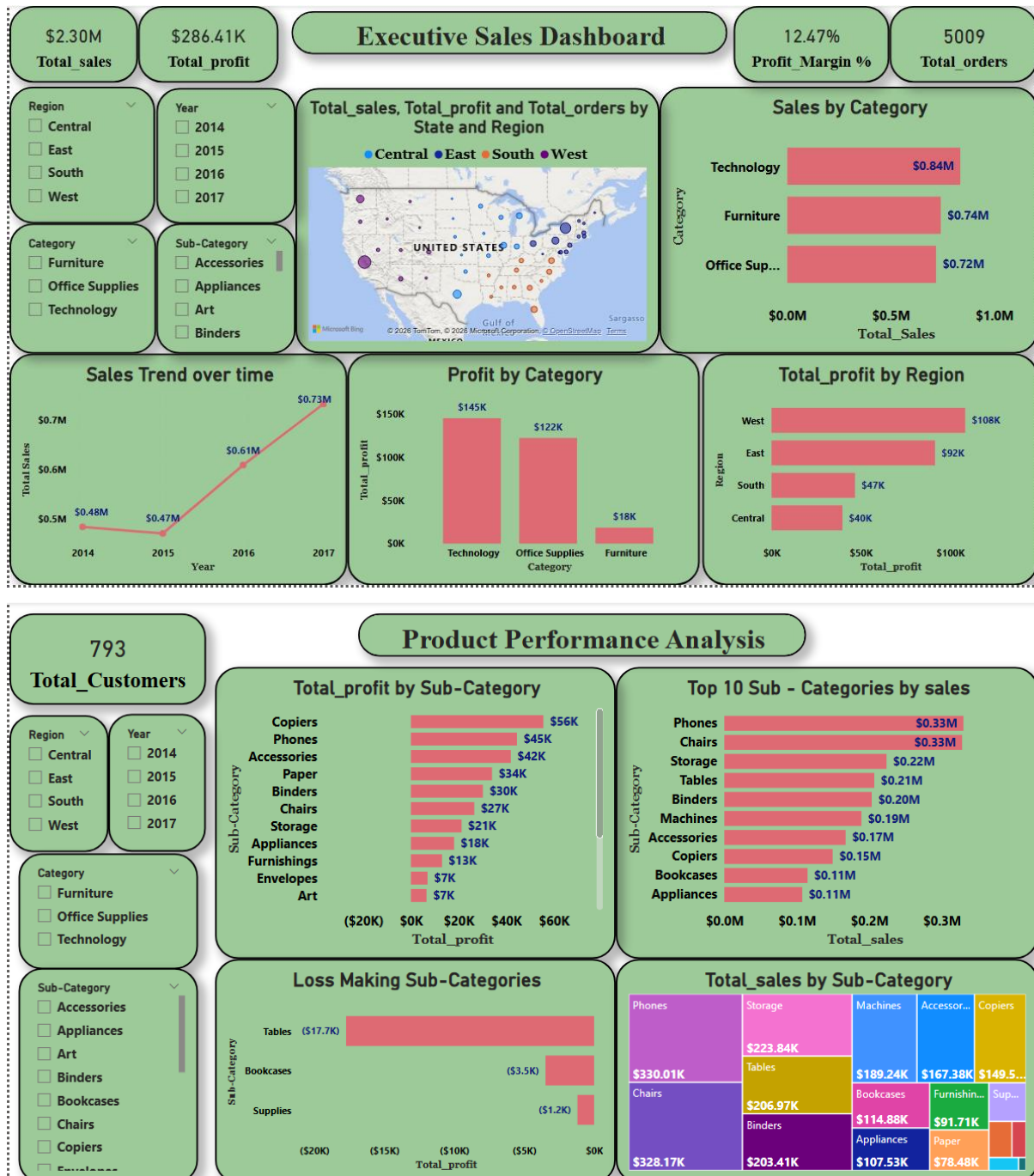
7. Data Export

- Exported the cleaned dataset into CSV format for further visualization and dashboard development in Power BI.

7. DAX (Power BI)

- DAX Measures: Implemented DAX formulas for key metrics, such as
 1. Total Sales = `SUM(Sales_Data[Sales])`
 2. Total Profit = `SUM(Sales_Data[Profit])`
 3. Total Orders = `DISTINCTCOUNT(Sales_Data[Order ID])`
 4. Total Customers = `DISTINCTCOUNT(Sales_Data[Customer ID])`
 5. Profit Margin % = `DIVIDE([Total Profit], [Total Sales], 0)`

8. Dashboard



9. Insights

1. Technology category generated the highest sales (USD 836K) and profit (USD 145K), making it the strongest performing business segment.

2. Sales showed a consistent upward trend from 2015 to 2017, with the highest sales recorded in 2017 (USD 733K), indicating steady business growth.
3. The West region contributed the highest sales (USD 725K) and profit (USD 108K), making it the most profitable geographical market.
4. Phones, Chairs, and Storage emerged as the top revenue-generating sub-categories, significantly contributing to overall business performance.
5. Tables, Bookcases, and Supplies were identified as loss-making sub-categories, with Tables generating the highest loss (USD -17.7K).

10. Recommendations:

1. Increase investment and marketing efforts in Technology products to maximize revenue and profitability.
2. Focus on expanding operations and customer acquisition strategies in the West region while replicating its successful practices in other regions.
3. Review pricing strategies, discount policies, and supply chain costs for Tables and Bookcases to reduce losses.
4. Optimize Furniture category profitability through better cost management and discount control, as it generates high sales but relatively low profit.
5. Maintain sufficient inventory and promotional campaigns for high-performing products such as Phones, Chairs, and Storage to sustain business growth.

11. Conclusion

The business demonstrates strong growth potential with increasing sales trends and high-performing Technology products. However, loss-making sub-categories such as Tables and Bookcases require strategic attention. Implementing data-driven decisions can help improve profitability and support sustainable business growth.